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13 May 1953
SUPERSEDING
ARMY 51-72-71
26 January 1949

MILITARY SPECIFICATION

PERISCOPE, 10.5

This specification consists of this cover sheet, Specification 51-72-71, dated 26 January 1949 and Amendment 2, dated 5 April 1949, attached hereto, without modification.

When a request for this specification is received by a supplylng activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifcations required by contractors in connection uith specific procurement functions should be obtained from the procuring ageng or as directed by the contracting officer.

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No. 51-72-71
26 January 1949
Superceding
Frankford Arsenal Tentative
FXS 968
10 December 1948

U. S. ARMY SPECIFICATION

PERISCOPE, M15

1. SCOPE AND CLASSIFICATION.

1.1 This specification covers one type of instrument known as Periscope, M15.

2. APPLICABLE DRAWINGS AND SPECIFICATIONS.

2.1 Ordnance Department drawing Class 22, Division 425, DRAWING 1, titled, Periscope, M15, List of Drawing, lists or indicates all drawings and specifications which form part of this specification.

2.2 All drawings and specifications shall be of the issue in effect on date of invitation for bids. The issue in effect shall be that issue which the contracting officer shall designate or furnish the bidder.

3. REQUIREMENTS.

3.1 Material shall be in accordance with the drarings, general specifications, and material specifications forming part of this specification.

3.2 Workmanship shall be in accordance with U. S. Army Specification No. 51-7 Instruments, Fire Control; General Specification Governing the Manufacture and Inspection of; and U. S. Army Specification No. 51-70-2 Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of.

3.3 General Requirements and associated tests common to all optical fire control instruments are contained in U. S. Army Specifications No. 51-7 and 51-70-2 and are mandatory in so far as they are Applicable.

3.4 Image, Quality."

3.4.1 Seven power system.- The images shall be practically free from distortion, astigmatism, curvature of field, coma, and chromatic and serial aberrations for a radial distance of 2 degrees 30 minutes from the center of the field of view. Images showing my of the above defects in the area specified above shall be cause for rejection of the periscope, if judged objectionable by the inspector.

3.4.1.1 Astimagism at the center of the field shall not exceed 0.25 diopter.

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3.4.2 Infinity sight- The image shall be clear, well defined and free of objectionable imperfections. It shall resolve one half minute.

3.4.3 Image quality shall, When specified, be equal to or better than that of a periscope furnished or designated by the contracting officer,

3.5 Collimation (Seven power).-

3.5.1 When two parallel rays pass through the head prism and are projected into the two objectives respectively, the conjugate rays emerging from the two eyepieces shall be parallel within the following limits at the infinity focus of the eyepiece and for any setting of the Interpupillary distance scale between 58 and 72 mm as follows:

Fourteen minutes of arc (2.0 minutes of true field) in the direction perpendicular to the plane determined by the two entrant rays.

They shall not diverge sore than twenty-eight minutes of are (4.0 minutes of true field), nor converge more than fourteen minutes of arc. (2.0 minutes of true field) in the plane parallel to a plane formed by the two entrant rays.

3.5.2 The line of sight through the center vertical reticle line shall be perpendicular within 3 mils to the axis joining the two mounting holes.

3.6 Image tilt (seven power) .- The image of a plumb line formed by the two eyepieces and as checked by an auxillary collimating telescope adjacent to each eyepiece shall not exceed the following tolerances:

45 minutes of arc of verticality
(2.3 nil when measured across 10° field)
30 minutes of parallelism
(1.5 mil when measured across 10° field)

3.7 Parallax.- When viewing a target at 75° plus or minus 50 yards, there shall be no parallax between the image and the reticle at the optical axis.

3.8 Plumb of reticle .- With the periscope vertical, with the axis joining the two mounting holes in a horizontal plane and sighting through the periscope: The broken reticle line shall not deviate from a horizontal target line by more than the angle subtended 0.4 mil at the extremity of the horizonal reticle line, when the other extremity of the reticle line is in coincidence with the horizontal target line.

3.9 Reticle accuracy .- The angular value between the zero point and any other graduation shall be as indicated within one percent plus 0.25 mil.

3.10 Eypiece focus , - The diopter scale index shall indicate zero within plus or minus 0.25 diopter when the reticle is brought into sharp focus with the aid of a dioptometer of at least 3 power or an equivalent auxiliary telescope. From this sharp focus the eyepiece shall have at least 2.5 diopters movement in both the plus and minus direction.

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3.10.1 With the same diopter setting on each eyepiece, one eyepiece shall not extend beyond the other by more than 1/32 inch.

3.11 Interchangeability of heads.-

3.11.1 Periscope heads shall be interchangeable and properly positioned the line of sight shall not diviate by more than a radial distance of 2 mils from the setting obtained with any other head.

3.11.2 With one periscope adjusted to meet the collimation and parallax requirements of paragraphs 3.5 and 3.7, the parallax between any interchanged head and the graduation mark nearest the optical axis (geometrix axis of reticle) and the image of a target at 750 ± 50 yards shall not exceed 1/8 diopter.

3.12 Interpupillary distance indicated by the scale shall be correct to within ± 1 mm.

3.13 Reticle illumination.- With an M33 type instrument light or equivalent mounted at the reticle window, the reticle shall appear properly Illuminated when tested in a dark room,

3.14 Temperature requirements specified in U. S. Army Specifications No. 51-7 for type I instrument shall apply.

3.15 Vitiation. - The periscope shall be subjected to the vibration test of U. S. Army Specification No. 51-70-2. The periscope shall not be damaged and it shall meet the collimation requirements.

3.16 Weatherproofing.- The periscope shall be properly sealed to prevent the entrance of rain into its interior.

4. SAMPLING, INSPECTION AND TEST PROCEDURE.

4.1 Unless otherwise specified each periscope and each spare part and assembly shall be inspected for conformce with this specification and the applicable requirements of U. S. Army Specification No. 51-7 and No. 51-70-2

4.2 Rain Test.- One periscope in every 25 shall be subjected to the rain test as Specified in U. S. Army Specification No. 51-7.

5. PREPARATION FOR DELIVERY.

5.1 Packing of instruments shall be in accordance with U. S. Army Specification No. 53-7 and supplementary instructions supplied by the contracting officer.

5.2 packing Of spare parts shall be in accordance with 0rdance Departmeat Packaging Instructions NO. AFPS-3 and/or AFPS-41-1.

5.3 Marking for shipment shall be in accordance with U. S. Army Specification No. 100-2 l

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6. NOTES:

6.1 The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

6.2 Notice,-This specification, together with specications and drawings pertaining to it and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procurement by the United States Government and shall not be reproduced either wholly or In part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Note: Copies of this specification may be obtained from the Office, Chief of Ordance, Department of the Army, Washington, 25, D. C l

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(No. 51-72-71)

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U. S. ARMY
SPECIFICATION

No. 51-72-71

Amendment 2
5 April 1949
Superseding
Amendment 1

PERISCOPE, M15

The following Change 9 shall be made in U. S. Army Specification No. 51-72-71, dated 26 January 1949:

Page 1, Paragraph 3.4.1. and 3.4.1.1 Delete and substitute the following:

"3.4.1 Seven Power System. - Resolving power on the axis of both telescopic systems for a parallel bar test object shall be 3,5 seconds of arc, or betters."

Page 2, paragraph 3.4.3. Delete.

Page 2, paragraph 3.7. Delete and substitute the following:
"3.7. When viewing a target at 750 plus or minus 50 yards with the aid of an auxiliary telescope of at least 3 power focused at infinity, the eyepieces shall indicate zero diopter within l/4 diopter."

Page 2, add the following paragraph.

"3.7.1. The dioptric difference between the Image of the target and the reticle graduation nearest the optical axis shall be between zero and minus 0.3 diopter when measured on the eyepiece scale."

NOTE: This condition must be met when using a head prism having no spherical power, or by adjusting for parallax without the head prism. Correction of fault indicated by failure of image to focus when eyepiece is within l/4 diopter of zero must be made by focusing objective.

* Page 2, paragraph 3.10. Delete.

* Page 3, paragraph 3.10.1 Renumber this paragraph to read 3.10.

* Page 3, paragraph 3.11.2. Delete & SUbstitute the following:

"3.11.2. With a periscope adjusted to meet the Prallax requirements of paragrpah 3.7, the parallax between the reticle graduation nearest the optical axis and the image target at 750 plus or minus 50 yards, introduced by Interchange of heads, shall not exceed diopter."

Page 3, add the following paragraph 4.3.

"4.3. Image Quality (Seven Power) -The test object shall be of the pattern illustrated in U. S. Army Specification No. 51-70-2, figure VI-A. The image shall be viewied with the aid of an 8 to 10 Power auxilary telescope of approved design, scaled in diopters. The test pattern shall be resolved in each of the four meridians by an adjustment of the auxilary telescope not to exceed ± l/8 diopter from the setting for best focus on the numeral in the center of the test field."

*Asterisks indicate changes from previous amendments.

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